

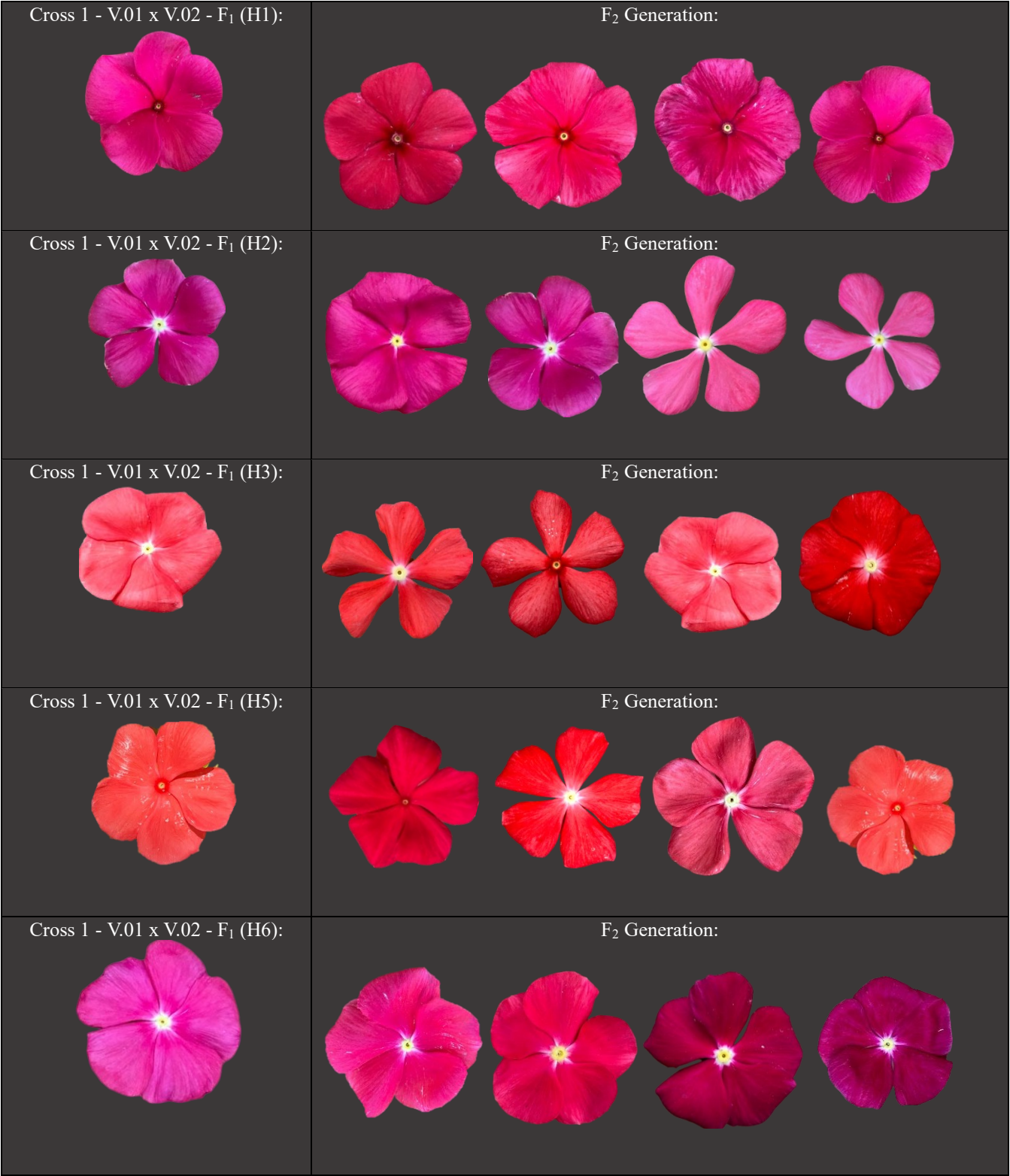
Supplementary Table S1. Morphological and agronomic characteristics of the parental genotypes of *Catharanthus roseus* used for hybridization and development of the F₂ populations.

Parental	Flower: main color	Flower color of receptacle	Flower: eye zone	Flower: arrangement of petals	Plant: growth habit	Leaf density
ORN UFPI V.01	light red	yellow	present	strongly overlapping	semi-erect	high
ORN UFPI V.02	dark pink	red	absent	strongly overlapping	semi-erect	high
ORN UFPI V.03	pink	yellow	present	touching	erect	intermediate
ORN UFPI V.04	white	red	present	touching	prostrate	intermediate
ORN UFPI V.05	magenta	pink	present	touching	erect	intermediate
ORN UFPI V.06	white	yellow	absent	free	erect	intermediate
ORN UFPI V.07	magenta	pink	absent	free	erect	intermediate

Supplementary Table S2. Number of F₁ plants obtained from crosses among *Catharanthus roseus* accessions from the Ornamental Plant Germplasm Collection of the Federal University of Piauí (UFPI), Teresina, Brazil.

Cross	Parental combination	Number of F ₁ plants obtained
Cross 1	ORN UFPI V.01 × ORN UFPI V.02	25
Cross 2	ORN UFPI V.03 × ORN UFPI V.04	18
Cross 3	ORN UFPI V.05 × ORN UFPI V.01	20
Cross 4	ORN UFPI V.06 × ORN UFPI V.05	40
Cross 5	ORN UFPI V.07 × ORN UFPI V.06	36

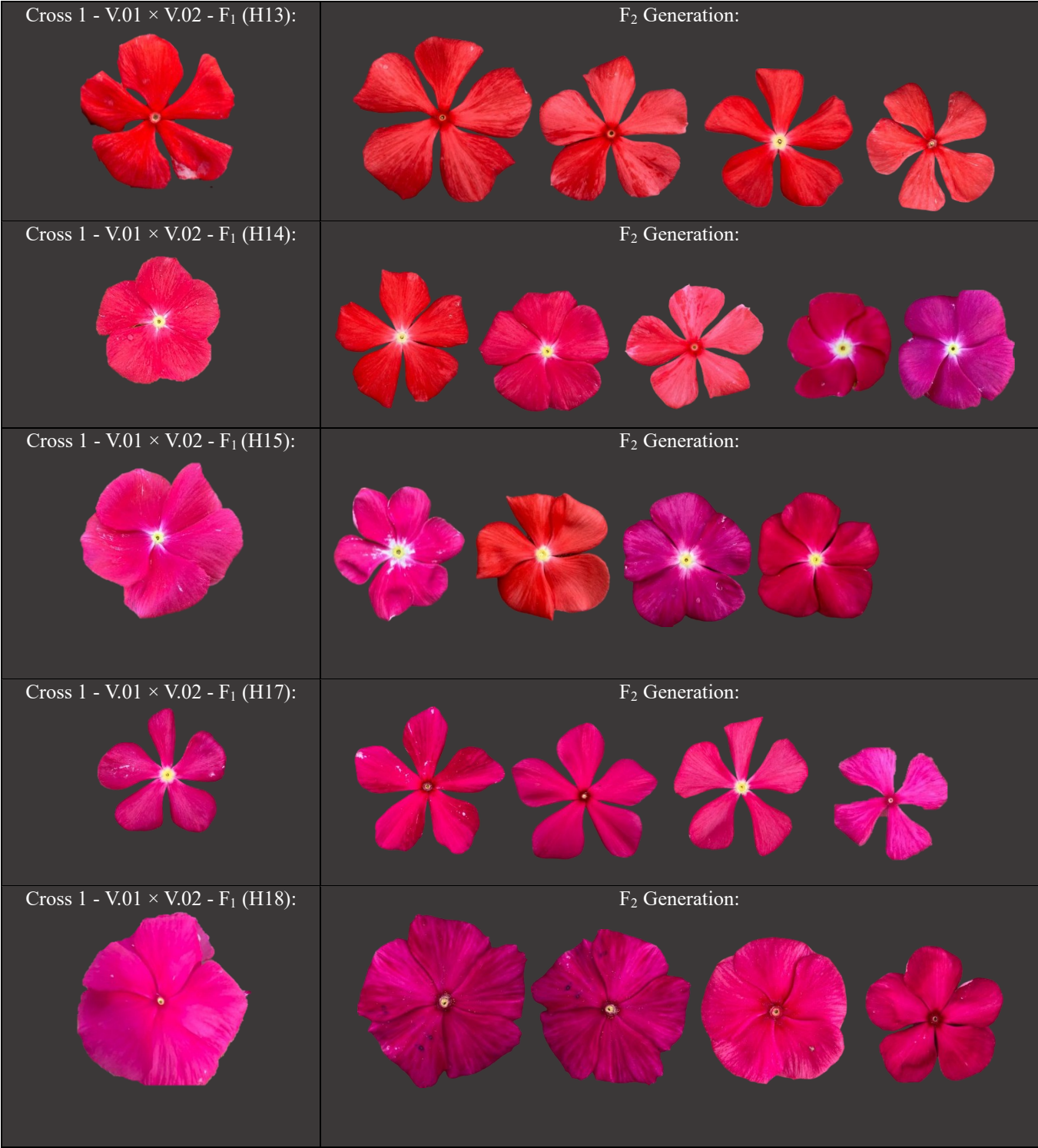
Supplementary Figure S1. Phenotypic variability observed in the F₂ progenies obtained through self-pollination of promising F₁ hybrids of vinca (*Catharanthus roseus* (L.) G. Don) (Continues).



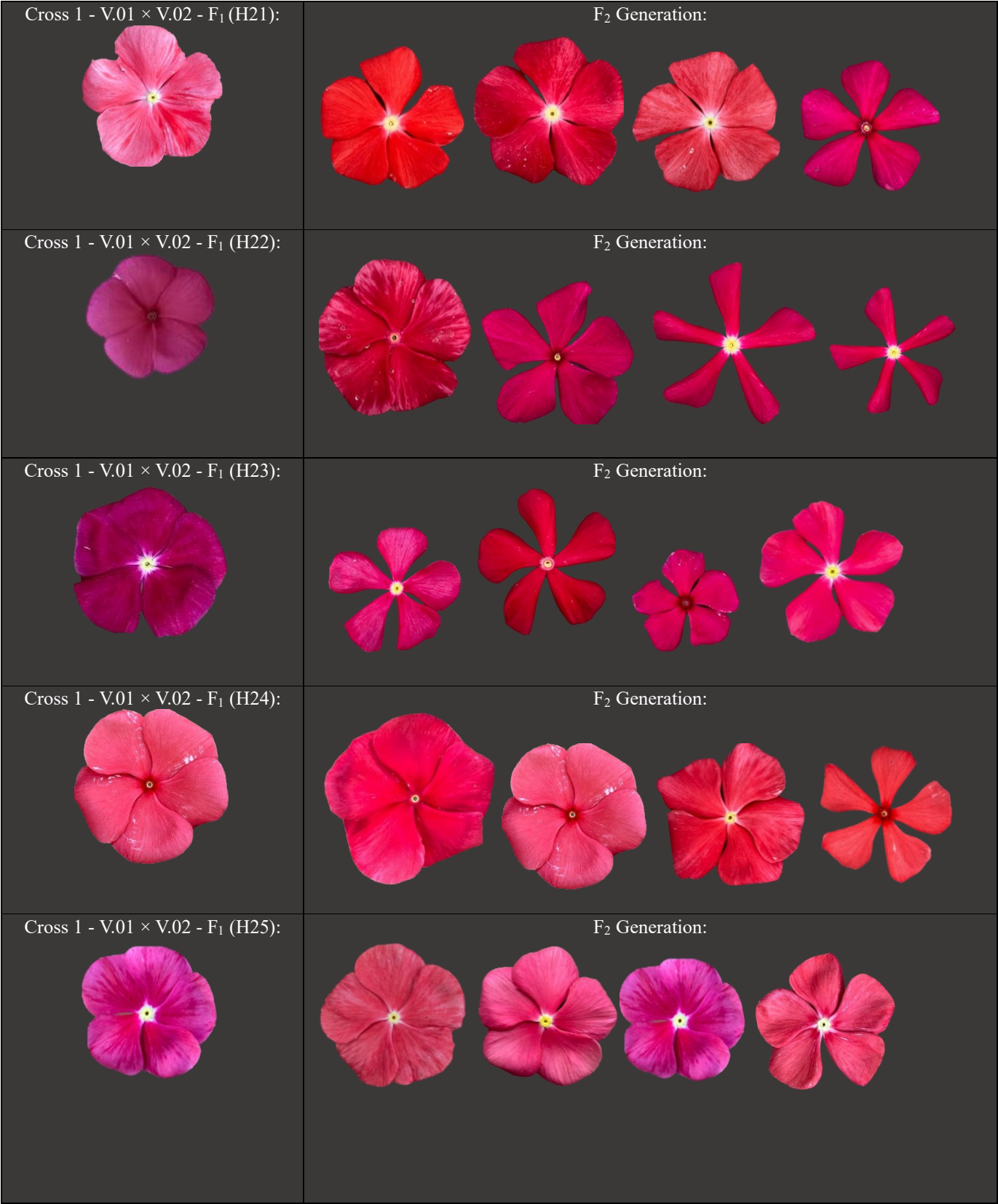
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Cross 1 - V.01 × V.02 - F ₁ (H8): 	F ₂ Generation: 
Cross 1 - V.01 × V.02 - F ₁ (H9): 	F ₂ Generation: 
Cross 1 - V.01 × V.02 - F ₁ (H10): 	F ₂ Generation: 
Cross 1 - V.01 × V.02 - F ₁ (H11): 	Geração F ₂ : 
Cross 1 - V.01 × V.02 - F ₁ (H12): 	F ₂ Generation: 

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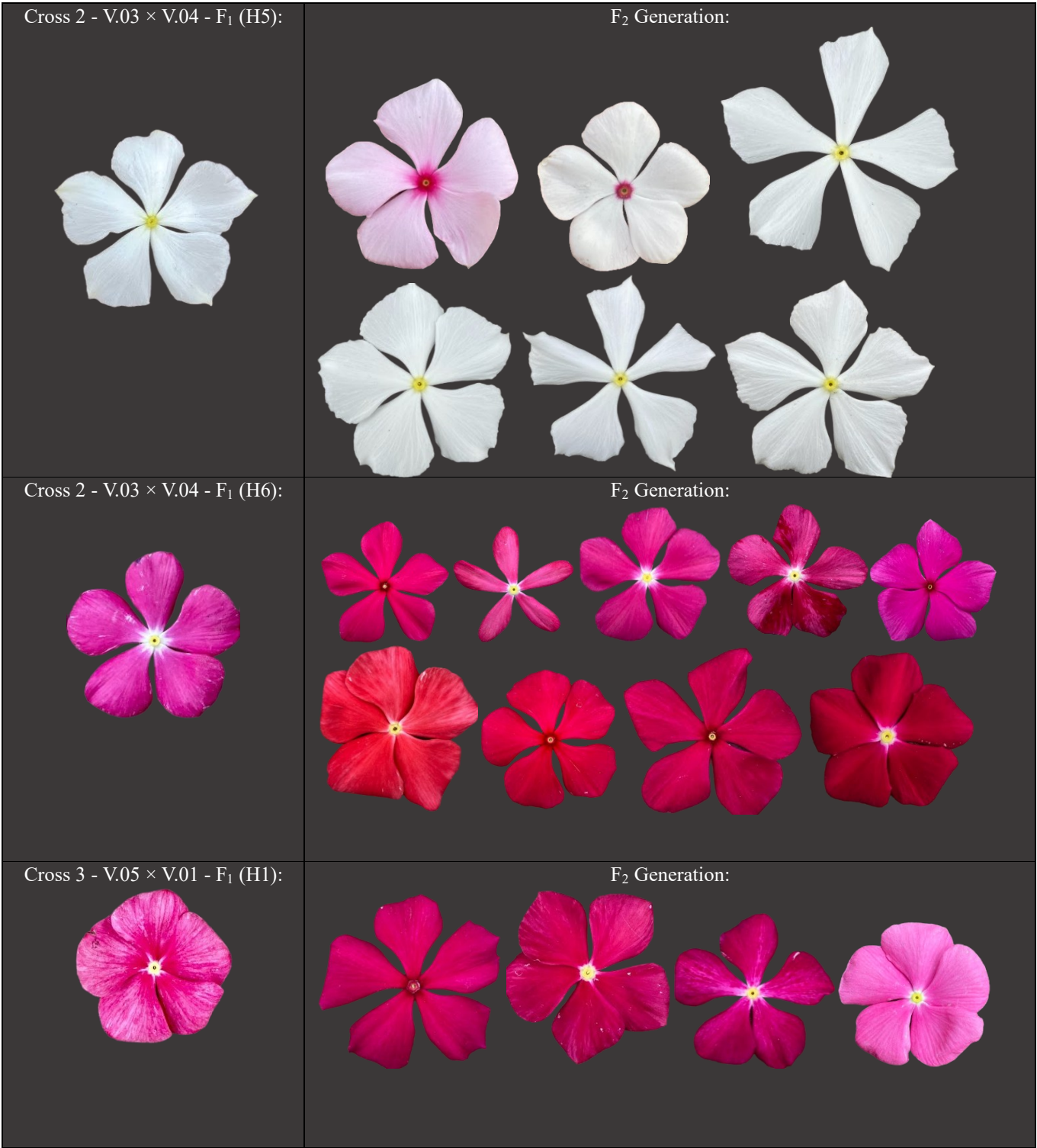
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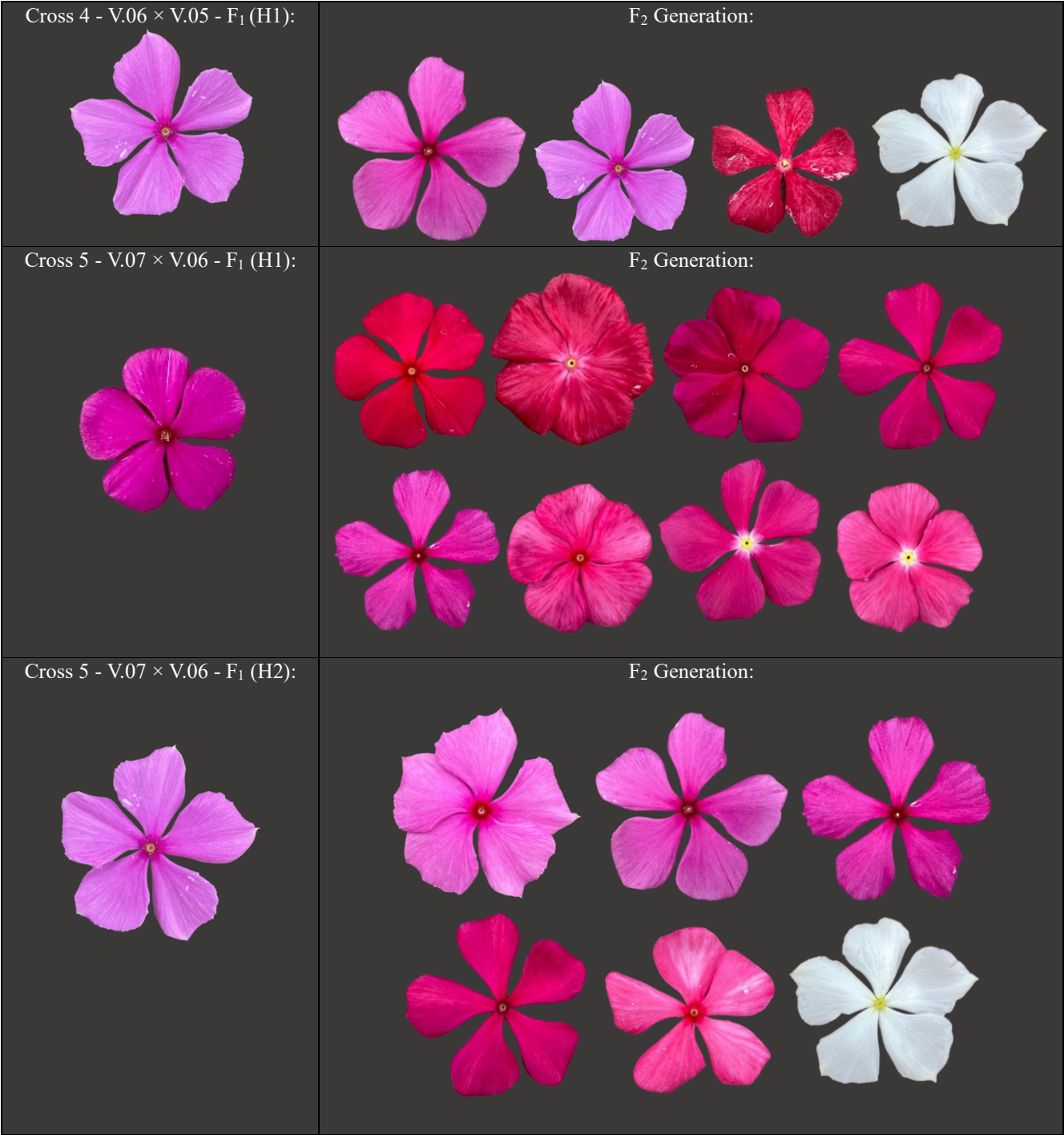
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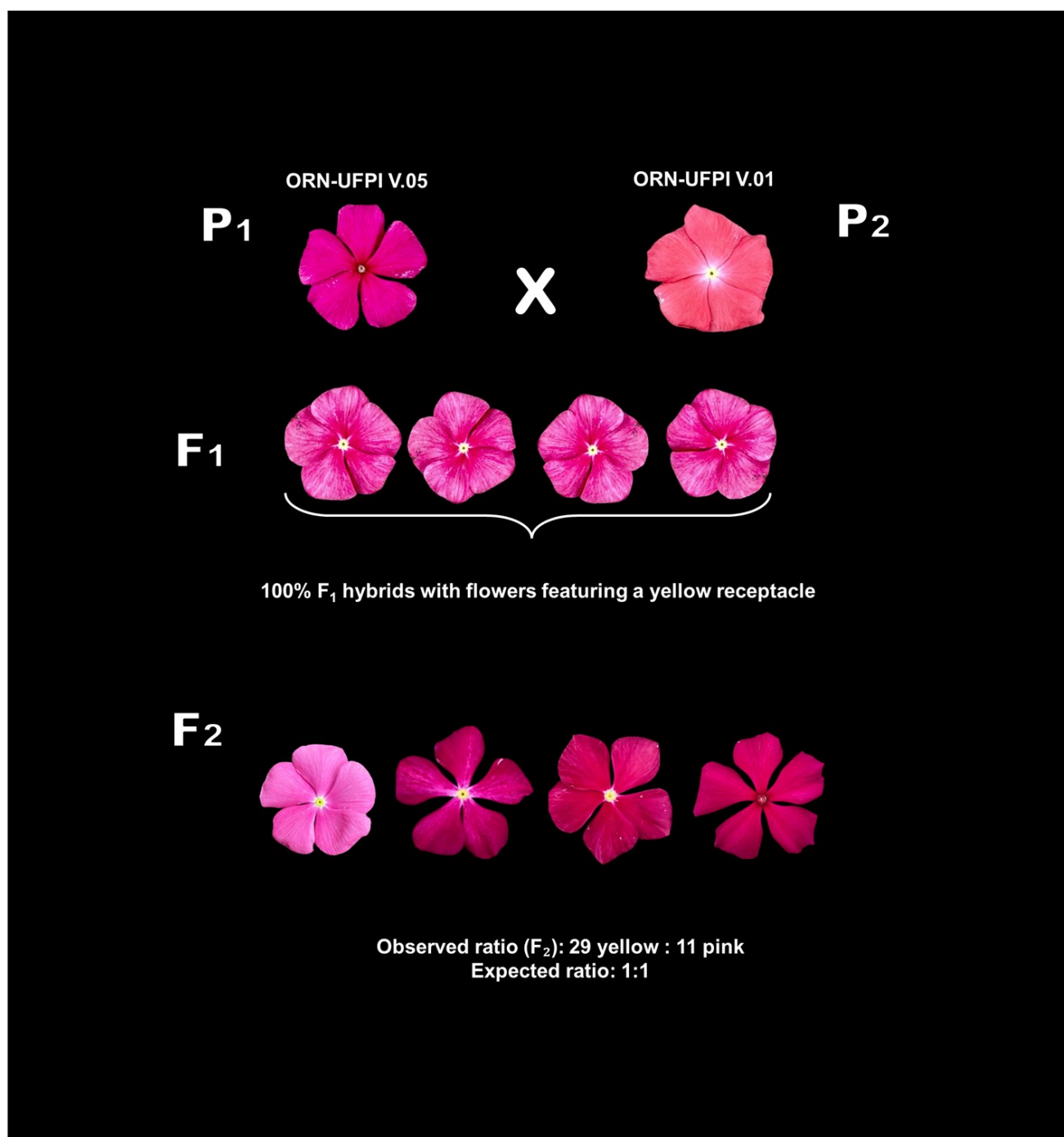
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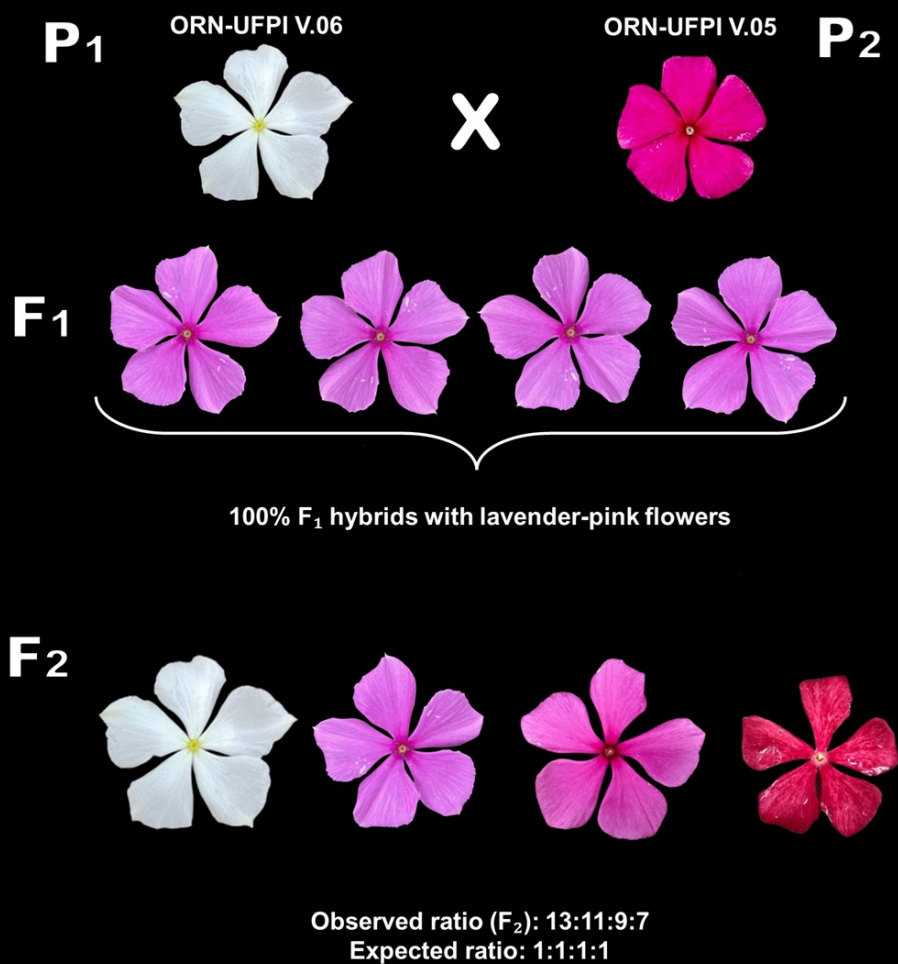
Supplementary Figure S1. Phenotypic variability observed in the F₂ progenies obtained through self-pollination of promising F₁ hybrids of vinca (*Catharanthus roseus* (L.) G. Don) (Conclusion).



Supplementary Figure S2. Phenotypic segregation for receptacle color in the cross 3 [ORN – UFPI V.05] × [ORN – UFPI V.01], showing the parental (P₁ and P₂), F₁ hybrids, and the observed segregation pattern in the F₂ generation.



Supplementary Figure S3. Phenotypic segregation of main corolla color in Cross 4 [ORN – UFPI V.06] × [ORN – UFPI V.05], showing the parental lines (P₁ and P₂), F₁ hybrids, and the observed segregation pattern in the F₂ generation.



Supplementary Figure S4. Phenotypic segregation of main corolla color in Cross 5 [ORN – UFPI V.07] $\times$ [ORN – UFPI V.06], showing the parental lines (P₁ and P₂), F₁ hybrids, and the observed segregation pattern in the F₂ generation.

